The Third Law of Thermodynamics Nobody knows what entropy really is, so in any discussion you
[width=0.8] /PhyAI/chapter3/3_1*.png* 2 possible picture arrangements There are many possible ways of sticking pictures; I
[width=0.7] /PhyAI/chapter3/3_2*.png* Large system has more entropy than the small system You can also increase the tota

Now imagine you are sticking pictures on your wall A, nearby is my wall B. I ask if you could decorate my wall too, so our
[width=0.6] /PhyAI/chapter3/3_3*.png* The total entropy of wall $A+B$ system When you start moving pictures to my wall

Inductively, we imagine the universe as a giant closed system, with many particles inside; and they are always exchanging "

An ice cube is more "organized" than a cup of water, and vice versa. In the middle of winter, water is frozen. Does this phe
[width=0.75] /PhyAI/chapter3/3_4*.png* Water is frozen Surprisingly, the answer is no. The water is cooled down since it **co**

But:

Entropy is closely related to energy, but they are not the same, and we do not have any law of conservation of entropy; so w

So the total entropy of our system (a cup of water and the outside environment) increases, because it gains more than losse

And obviously, also by the law:

The temperature itself is not so important than the **difference**. If our system only consists objects at same temperature, n

In this section, we develop the connection among three physical quantities: (**heat**, **energy**, **entropy**). Let's begin with a ve

Plot our data into a simple graph: [H]
[width=0.7] /PhyAI/chapter3/3_5*.png* A simple graph of the entropy Looking at table and graph, we find out some interest

Again, the total entropy is increasing! Can I guess the slope of a tangent line at point $q_A = 20$ is **steeper** than point $q_B =$

Substituting our values yields:

Here is a question, why is ΔS_B negative, but it still has positive slope? The answer is pretty simple, our graph perspective

According the zeroth law of thermodynamics, the temperature of solid A and B are equal if and only if they are in equilibr

A subtle observation is that a decrease in q_A corresponds to an increase in q_B:

This equation makes perfect sense to our intuition, look at previous calculation: the slope of S_A graph is much steeper than

To ensure the second law of thermodynamics, as you can see here, the entropy of solid A increases **faster** than the rate of e

Apparently, we can see that: